

The Impact of Environment Design Bias on Working Memory

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(a) A view of the stereotypical classroom.



(b) A view of the neutral classroom.

Fig. 1: (a) The self-avatar is seated in the center of the room. The Einstein poster and the “Stay Focused” poster of two men on a construction beam were swapped to different images in the neutral environment. (b) The national parks poster and the growth mindset poster were swapped to different posters in the stereotypical environment. All other posters in both environments remained the same.

Abstract— Extended Reality (XR) is a powerful tool for training, education, and gaming. Research suggests that gender differences exist in XR environments including women having a lower sense of subjective presence and being more susceptible to motion sickness. However, the underrepresentation of women both as participants and researchers could lead to potential design biases, impacting the accuracy and inclusivity of XR systems. This work investigates subtle design differences in virtual environments on women’s performance on a cognitive test. Non-male participants ($n = 40$) completed the Stroop Interference Task in two virtual classroom environments: a neutral and a stereotypically STEM environment. The environments were altered by four wall posters depicting positive gender-neutral and nature posters to science-fiction and positive male figures, such as Albert Einstein. Results support that when participants were in the stereotypical environment they were more distracted and responded more slowly and less accurately than when they were in the neutral environment. Additionally, positive female self-avatars buffered participants from the negative impacts of the stereotypical environment. These results highlight the need for more inclusive research practices. Minor adjustments can significantly improve or harm women’s engagement and performance in XR settings. We emphasize the importance of bias awareness in study design, and recommend that researchers consider how their experiments could impact participants of all demographics, in order to enhance inclusivity and non-biased results.

Index Terms—Environment design, study design. design bias, gender bias, avatars, avatar embodiment.

1 INTRODUCTION

Extended Reality (XR) is considered a powerful tool that supports investigation of behavioral sciences [3], enhances training [20], and fosters education [21]. Within XR research, numerous studies have identified gender differences. For example, results suggest that women have a lower sense of subjective presence than men in XR [13] and are more susceptible to simulator sickness [15]. These results support that XR may be “sexist” [39] and that women may have worse XR experiences compared to men.

The differences in XR experiences between men and women may be caused by experimental design. This work investigates if subtle design decisions in a virtual environment (VE) can impact women’s performance on cognitive tests. Current negative experiences, such as increased simulator sickness or feeling less present, may partially explain why women are underrepresented in XR research as both par-

ticipants and researchers [33]. Of note, numerous calls have been made to increase representation in XR [31, 39] including recommendations for how to support diversification. However, the current state of the field is male dominated with systems and experiments being designed by, tested on, and run by majority men [31]. Though unintentional, a homogeneous research population is more likely to add design bias into their systems. Design bias is defined as bias in results that can be attributed to the study or implementation design [37]. One example of the negative effects of design bias is facial recognition systems being less accurate with Black faces and female faces compared to White male faces [5].

When considering XR human subject research experiments, researchers make design decisions about the tasks participants complete during the experiment and the environments where participants are immersed. These experimenter decisions may inadvertently add unintentional bias into the experiment that unknowingly impacts groups of participants, resulting in design bias.

Small changes to VEs have been shown to influence user behavior. Research by Cheryan et al. [7] found that changing the posters in a virtual Computer Science classroom from stereotypically Science, Technology, Engineering, and Math (STEM) to neutral increased women’s intentions to enroll in Computer Science courses. This simple design change of posters on a wall affected women’s sense of belonging in the space. Further, stereotypical STEM images, including male role-models or science fiction, may have unintended exclusionary effects on

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women [9]. However, it is unclear if small changes, such as changing the posters on a wall, impact cognitive performance in VEs.

In this work, we focus on non-male participants, since both women and non-gender-binary individuals have a similar lack of confidence in STEM fields compared to male-identifying students [19]. For the remainder of this paper we will refer to non-male identifying individuals as women+. We investigate if subtle XR design decisions from neutral environments compared to stereotypically STEM environments negatively impact performance for women+. To test this, we conducted a mixed-design VR user-study on women+ participants in both a neutral and stereotypical STEM environment. We investigated the impact of the environment on user performance on a standard cognitive test, the Stroop Interference Task, as well as subjective sense of feeling included in the environments.

2 BACKGROUND

2.1 Gender and Virtual Reality

Numerous XR studies have identified gender differences related to the subjective sense of presence, emotional response, and performance. When considering presence Schwind et al. [38] found that virtual hand representation impacted presence differently for men and women. Women with male hands had lower presence and likability scores, while reporting higher levels of eeriness and had higher presence scores when using non-human hands, such as robotic or abstract cartoon hands. On the other hand, men felt more presence when using any human hand, as well as female hands. This suggests that virtual embodiment research experiments that did not provide participants with inclusive hand options may have negatively impacted women's subjective sense of presence. Since higher presence is linked to learning [16], these experiments have have inadvertently hindered women's performance.

Previous work has also identified that women may have a lower sense of subjective presence in XR compared to men [13]. In this work, presence was measured after participants gave a virtual presentation on the Kingdom of Bhutan. At a high level, the topic of the Kingdom of Bhutan may appear innocuous. However, when considering gender, the Kingdom of Bhutan is a monarchy recently governed by a king who practiced polygamy and had four wives. A presentation on a monarchy that supports polygamy could have impacted women's feeling of inclusion in the experiment, potentially influencing responses, and possibly leading to unintentional design bias.

When considering emotional responses, multiple studies have found that women may have more positive emotional responses to Augmented Reality (AR) and VR programs than men [10, 25]. Dirin et al [10] had participants watch advertising videos and identify a cute cat displayed in AR and found that female participants reported more positive emotions than male participants. However, gender differences have been identified in emotion expression such that women are more likely to express positive emotions compared to men [6]. Further, Mousas et al. [25] found that women had significantly more negative emotional intensity to virtual characters compared to men. However, the virtual characters investigated in the experiment only included male characters. Including female appearing characters may have produced different results.

There is contradicting evidence on simulator sickness and gender. Many studies have found that women are more susceptible to simulator sickness compared to men [15]. However, research by MacArthur et al. [22] who performed a systematic review of simulator sickness studies identified confounding factors, including experimental tasks and content, that may have led to design bias. This suggests that previous simulator sickness experiments should be carefully reviewed to determine if experimental bias could have influenced results.

Related to the present work considering performance on learning tasks in VR, previous results found that male participants outperformed female participants on a learning task [1]. However, in that work, the learning task was a military based exercise where women have historically been excluded possibly biasing results. Conversely, Suma et al. [43] found that women outperformed men on more complicated navigation tasks.

In this work, we investigate if subtle environment design choices can impact women's performance in a VE. Identifying differences in performance would highlight the need to investigate possible design bias from previous studies, such as some of the studies listed above, that identified gender differences.

2.2 STEM Stereotypes and Environment

Gender stereotypes about STEM and Computer Science start in children's education as early as age six. These stereotypes may continue throughout adolescence and have been linked to the contribution of gender disparities in STEM fields [24]. A higher belief in stereotypes that favor men in STEM fields has been shown to be associated with lower levels of belonging and interest for women in STEM [24]. Sterling et al. [42] also found a gap in confidence between men and women in STEM such that women have lower confidence in their STEM abilities compared to men.

When considering environments, [8] Cheryan et al. found that female undergraduates expressed less interest in Computer Science when in a real classroom environment with stereotypical objects associated with masculinity, such as Star Trek or video games posters. However, when the environments were redesigned to include more neutral objects, such as art or water bottles, women's interest in Computer Science increased to comparable levels to men's.

These results were replicated in VR by Cheryan et al. [7] who examined whether the design of a virtual learning environment influences a student's intentions to enroll in an intro level Computer Science class, and their anticipated success. Specifically, they were curious about how a stereotypical, masculine classroom design impacts students, particularly female students, compared to an inclusive, not stereotypical environment. They found that switching to a non-stereotypical environment increased women's interest in enrollment and their anticipated success, but did not impact men's interest in enrollment or anticipated success.

The objects in a physical room were shown to impact women's sense of belonging in a Computer Science classroom. In classroom settings with stereotypically masculine objects, such as science fiction posters, tech magazines, and video games, women were less interested than men in Computer Science. Conversely, there were similar levels of interest for men and women in a pre-measure of interest, as well as in a non-stereotypical environment that did not contain these items [23]. Women also were less likely to report that they fit the stereotypical image of a computer scientist and were less likely to enroll in a stereotypical classroom because of this.

Master et. al [23] studied the impact of environment on women's belonging and interest in STEM. Women vastly preferred an inclusive environment over a stereotypical environment, and were much more likely to take a class in an environment with non-stereotypical objects such as nature, art, and plants. Women also felt a higher sense of belonging from the non-stereotypical environment. Other research into classroom environments and belonging shows that students who feel as if they do not fit into the classroom experience detrimental effects on their learning [36].

This previous work demonstrated that the environment, both real and virtual, impacted women's sense of belonging and likelihood to continue taking Computer Science courses. However, it is unclear if a stereotypical STEM environment would negatively impact women's cognitive performance in VEs.

2.3 Avatar Embodiment

Self-avatars have been known to improve performance on cognitive tests [34, 40] and improve distance estimation [35]. Further, Yee et al. [45] found that people's behavior stereotypically conforms to the appearance of their self-avatar, known as the Proteus effect. For example, people embodied in attractive avatars were more intimate in self-disclosure tasks compared to being embodied in less attractive avatars. Additionally, people embodied in taller avatars behaved more confidently in negotiation tasks compared to people embodied in shorter avatars. Additional examples of the Proteus effect include light-



Fig. 2: In the Neutral environment Avatar condition, a view of a self-avatar looking at their reflection in the mirror. Visual feedback about color selection was provided to the user. Two white circles, mimicking the trackpad were always visible to the user. As the finger touched the trackpad on the left, up, or right, the corresponding selected color would appear on the screen. In the image above, the user's thumb is touching the top of the right touchpad to select the green color.

skinned people having reduced implicit racial bias when embodying dark skinned avatars [32].

Related to this work, participants embodying an avatar of Albert Einstein increased their performance on the Tower of London cognitive task [2]. Further, women embodying male avatars under stereotype threatening situations were buffered from stereotype threat and did not experience performance impairment on a working memory task [28,30].

3 METHODS

The following mixed-design user study was approved by the Davidson College Human Subjects Institutional Review Board. This work investigated the following hypotheses.

H1: The environment will impact women+ performance. Specifically, women+ performance will be negatively impacted in the stereotypical environment compared to the neutral environment.

H2: The environment will impact women+ subjective sense of inclusion. Specifically, women+ sense of inclusion will be negatively impacted in the stereotypical environment compared to the neutral environment.

H3: Participant performance will be improved with a self-avatar compared to participants without a self-avatar.

H4: Participants with higher avatar embodiment will have better performance.

3.1 Measures

3.1.1 Stroop Interference Task

The Stroop Interference Task [17] is a common psychological test. Participants are shown written out colors such as “red” or “blue” in different font colors and are asked to identify the color of the font, not the color that is written. Participants are quicker and respond more accurately when the written color and the font color are congruent (e.g. “red” written in red). Alternatively, participants are slower to respond and are less accurate when the written word and the font color are incongruent, (e.g. “red” written in blue).

In this work we implement the Stroop Interference Task following the same guidelines presented in [34]. For each Stroop trial, text appeared presenting either “Red”, “Green”, or “Blue” and was written in either red, green, or blue font. The Stroop Interference Task requires participants to select the correct color of the displayed text, not what the text reads. Participants selected the color via the touch pad on the hand held controller, left-red, up-green, or right-blue. Visual feedback was provided to the user via two white circles in the scene that visualized the buttons being touched on the touch pad. When the user touched the

touch pad a small colored circle appeared in the display to indicate the color selection. The visual feedback is shown in Figure 2.

To make the test more challenging, following the procedure in [34], 80% of trials were congruent and 20% of trials were incongruent. This ratio of congruent to incongruent trials was used in order to maximize the difference in latency and to catch the participant off guard [18]. If participants took more than 5000ms, the trial automatically timed out and advanced to the next trial.

Performance measures include *accuracy* in answering the correctly written color, and *latency* to respond on correctly answered trials. Participants are often faster to respond and more accurate on congruent trials, and have lower accuracy and are slower to respond on incongruent trials. Performance on incongruent trials will have been shown to be negatively impacted (less accurate or slower to respond) when working-memory is impaired [18].

3.1.2 Head Yaw

Head yaw angle is a common measure of distraction [26]. Further, how much people turn their heads in a VE is an indication of how much they are looking around the environment. Head yaw can be an indication of how distracted participants may be while taking a cognitive test, or how much of the environment they observed. For example, Won et al. [44] considered head pose when investigating student anxiety levels in a classroom environment and found that the standard deviation of yaw head movement was correlated with anxiety for women.

In this work, we measure head yaw and consider the yaw standard deviation (SD) while taking a cognitive test. A high yaw SD would indicate that the participant is looking around more during the test and may therefore be more distracted.

In this experiment participants were additionally asked to look around the environment to become acclimated with the environment before taking the cognitive test. During this experiment phase we consider the percentage of time participants are at the yaw mode—the most frequent yaw angle—while looking around the scene. A low yaw mode would indicate that no single object in the scene was looked at for an extended amount of time. A high yaw mode would indicate that one location was looked at more frequently than other locations around the scene, such as staring at a poster in the room. Staring at stereotypical posters, as used in this experiment, may activate stereotype threat [41] which has been linked to slower response times on cognitive tests [27].

3.1.3 Questionnaires

Participants completed the Avatar Embodiment Questionnaire (AEQ) [29], a subjective measurement of how embodied people are in their self-avatar. All participants completed the AEQ even though half of participants did not have a self-avatar. Results were analyzed only for participants with a self-avatar.

To investigate if participants' sense of belonging replicated previous work [7,9], participants additionally were asked how much they agreed with the following statements on a 7-point Likert scale from completely disagree to completely agree:

Belong: I belonged in this environment.

Threat: I was threatened by this environment.

Confidence: The environment increased my confidence.

Succeed: I can succeed in this environment.

Participant responses to Belong, Threat, Confidence, and, Succeed, (with Threat negatively encoded) were summed and averaged into one measure, *Inclusion*, with acceptable reliability (Chronbach's $\alpha = .73$). Values for Inclusion were normalized on a scale from 0-1.

3.2 Participants

A total of 44 participants took part in the experiment. Participants were recruited through flyers hung up around campus and town. Eligibility requirements included: being 18 years or older, having normal or corrected to normal vision and hearing, being proficient in spoken English, not having a history of epilepsy, seizures or motion sickness, not identifying as a man, not being knowingly colorblind, and being

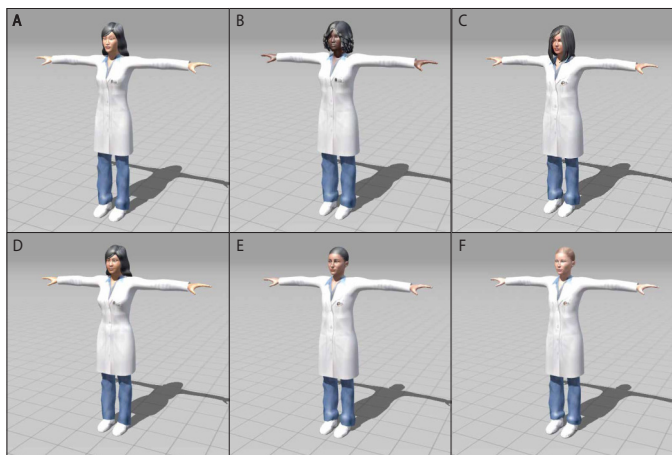


Fig. 3: The six racially diverse avatars from the VALID avatar collection representing A) East Asian, B) Black, C) Hispanic, D) Native Hawaiian/Pacific Islander, E) Middle Eastern/North African, and F) White. Participants in the Avatar condition selected the avatar that they most identified with.

able to comfortably use one or both hand-held controllers. Four participants were removed, two due to technical difficulties, and two who did not understanding the Stroop interference instructions. Of the 40 participants included in the experiment, 10 participants were in each of the four conditions. 37 participants identified as women and 3 identified as non-binary. Participant ages ranged from 19-51 years ($M = 24$ years, $SD = 9$ years). Participants self-identified their race as Black (2), East Asian (5), White (31), Hispanic or Latino (2), South Asian (3), and Other (1). The racial identification adds to over 40 since some participants identified as more than one race.

3.3 Equipment

Participants wore an HTC Vive Pro head-mounted display (HMD) with approximately 110° field-of-view. The experiment was rendered on an Alienware computer running on Windows 10 Pro edition with an Intel i7-7820X processor (3.6 GHz), 32GB RAM, and NVIDIA GeForce GTX1080 Ti GPU. The experiment was created in Unity.

3.4 Procedure

Participants were welcomed into the experiment room and guided to a small desk and computer in the far corner. On this computer, participants completed an informed consent form and confirmed their eligibility. Participants were randomly assigned into one of two between-participant conditions, **Avatar**: participants with a self-avatar and **No Avatar**: participants without a self-avatar. Participants in the Avatar condition were shown the faces of six avatars and were asked to select the avatar that they most identified with. Avatars were sourced from VALID avatars [12] and were all female appearing and wore a scientific lab coat. The avatars are shown in Figure 3. Participants in this condition were also asked for their height, and the avatar heights were scaled to each participant's height before the experiment began.

Participants were then guided to the center of the room where they were seated in a physical chair at a physical table. Participants were introduced to the experiment procedure and equipment. Participants were handed the hand-held controllers and an HTC Vive Pro HMD was placed on their head. Participants were instructed to keep their hands and hand-held controllers on their laps. The hand placement was selected to make the task more challenging by limiting processing capabilities [34], instead of on the table next to the experiment task. Additionally, participants were asked to keep their feet flat on the floor for the duration of the experiment. The experiment was then started and the participant found themselves in a virtual classroom.

The experimental procedure is visualized in Figure 4. Each participant completed two consecutive experiment blocks, **Neutral** and

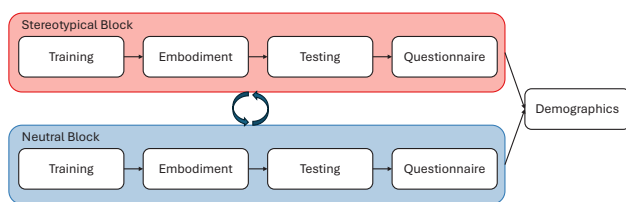


Fig. 4: A visualization of the experiment procedure. Each participant completed two blocks, stereotypical and neutral in counterbalanced order. Each block had four phases, training, embodiment, testing, and a questionnaire. Finally participants completed a demographics questionnaire. Participants completed the experiment either with an avatar or without an avatar for the entire experiment.

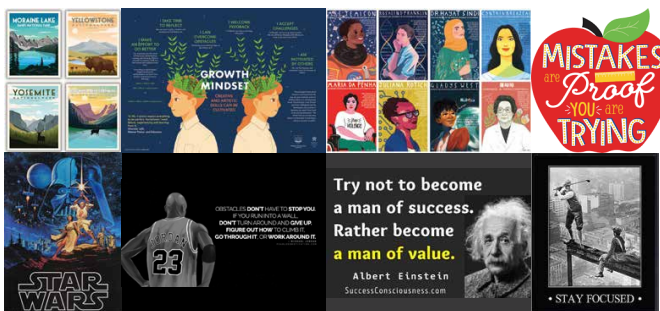


Fig. 5: Top row: four posters in the Neutral environment. Bottom row: four posters in the Stereotypical environment. Each column represents the poster that it was swapped with in each environment. All posters were positive or neutral in design. The posters in the stereotypical environment included images of men.

Stereotypical with order counterbalanced across all conditions. The experiment block procedure was identical and varied only by environment design. The experiment block included four phases: (1) training, (2) embodiment, (3) testing, and (4) questionnaire.

Both the Neutral and Stereotypical environments consisted of a classroom environment with six large tables each with a chair. A view of the stereotypical STEM and neutral virtual environments are shown in Figure 1. A whiteboard was placed in the front of the room. Each room had nine posters on the walls and the only differences between the environments were four of the nine posters on the walls. Five posters were held constant between both environments and included positive educational posters and neutral landscapes. In the Neutral environment, the remaining four posters (Figure 5 top row) included a neutral poster of the USA national parks, a gender inclusive image about growth mindset including a White man and woman, an inclusive image of racially and ability diverse STEM role models, and a neutral image about making mistakes. In the Stereotypical environment, (Figure 5 bottom row) the four posters listed above were swapped out with a stereotypically masculine Star Wars movie poster, a Michael Jordan quote, an Einstein quote, and an image of two white men on a construction beam. A Pilot study supported that the selected images were seen as either neutral or positive, and stereotypically STEM or not stereotypically STEM. No image was seen as negative or stereotypically feminine.

In the environment, participants were seated at the front middle desk facing the white board. The desk and chair were collocated with the real desk and chair to provide passive haptic feedback. A virtual mirror was placed on the desk. Participants in the Avatar condition were collocated with their self-selected avatar and could see the avatar's reflection in the mirror as shown in Figure 2. The avatar's head moved synchronously with the participant's head. Participants in the No Avatar condition did not have a self-avatar.

Participants listened to audio instructions, via headphones, that

Effect	Accuracy		Latency	
	F	p	F	p
Trial	0.37	.545	25.49 ***	<.001
Avatar	0.01	.934	15.60 ***	<.001
Env	0.66	.418	6.19 *	.014
Yaw	5.88 *	.017	0.65	.422
Trial:Avatar	0.17	.683	3.08 +	.083
Trial:Env	2.58	.111	0.63	.428
Avatar:Env	1.39	.240	0.98	.323
Trial:Yaw	5.12 *	.026	0.03	.854
Avatar:Yaw	4.02 *	.047	15.18 ***	<.001
Env:Yaw	3.44 +	.066	10.34 **	.002
Trial:Avatar:Env	0.21	.648	0.21	.648
Trial:Avatar:Yaw	4.81 *	.030	1.91	.170
Trial:Env:Yaw	5.28 *	.023	1.84	.178
Avatar:Env:Yaw	4.44 *	.037	0.67	.414
Trial:Avatar:Env:Yaw	1.11	.294	0.64	.426

Table 1: Output of the Mixed Model ANOVAs for Accuracy and Latency of Stroop performance. Env is abbreviated for Environment.

guided them through the experiment procedure. Participants first entered (1) the training phase to teach them how to complete the Stroop Interference Task. Audio instructions guided participants through 10 practice trials to ensure they understood the instructions. Feedback was given on practice trials such that if a mistake was made during these trials, a red “X” appeared until participants corrected their mistake. Participants were then guided through (2) the embodiment phase asking them to look around the environment, look down at their legs, and to look in the mirror. Participants were also asked to describe out loud what they saw around the environment. Qualitative data was recorded from their verbal responses. The embodiment phase took approximately two minutes.

The embodiment phase was immediately followed by (3) the testing phase. During the testing phase, participants completed 200 trials of the Stroop Interference Task. Participant accuracy and latency to respond was recorded for each trial. Additionally, 6 degree-of-freedom head tracking data was recorded throughout the experiment.

After 200 trials, the testing phase concluded and participants completed (4) the questionnaire phase outside of the VE. Participants exited the virtual environment, by taking off the HMD, and completed a questionnaire administered on a desktop computer. Participants first completed the Avatar Embodiment Questionnaire (AEQ) [29] and then the Likert-scale Inclusion questions.

After completing the first block, participants repeated the same procedure, completing the training, embodiment, testing, and questionnaire phases, in the other virtual environment. Half of the participants in the Avatar condition and half of the participants in the No Avatar condition completed the Stereotypical Block followed by the Neutral Block. The other half of participants completed the Neutral Block followed by the Stereotypical Block. Finally, participants completed a demographics survey and were debriefed on the purpose of the experiment. Participants completed the total experiment in approximately 45 minutes and were compensated in the form of a \$15 US gift card.

4 RESULTS

Analysis was performed in R, version 4.4.1.

4.1 Accuracy

Accuracy - the percentage of correct responses, was analyzed with a Mixed Model ANOVA with explanatory variables Trial (In-congruent, Congruent), Avatar (Avatar, No Avatar), Environment (Neutral, Stereotypical) and yaw standard deviation (SD) during the testing phase. The ANOVA output is reported in Table 1 (left).

A significant main effect of yaw SD was found, ($F(1, 114.38) = 5.88, p = .02, \eta^2 = .05$). Participants with lower yaw standard deviations had higher accuracy. That is, participants who did not move their heads as much were more accurate. This main effect is mediated by

multiple 2-way and 3-way interactions. Since the 2-way interactions are both mediated by the 3-way interactions, post-hoc analysis, using estimated marginal trends is reported on the 3-way interactions.

There was a significant Trial $\times$ Avatar $\times$ yaw SD 3-way interaction, ($F(1, 110.59) = 4.81, p = .03, \eta^2 = .03$). See Figure 6. During the harder incongruent trials, participants without a self-avatar had a significantly more negative yaw SD slope compared to participants with a self-avatar, ($t(101.5) = 2.54, p = .01, r = .24$). Participants in the No Avatar condition with more head movement had lower accuracy during the incongruent trials compared to participants with less head movement. The same was not seen for participants in the Avatar condition.

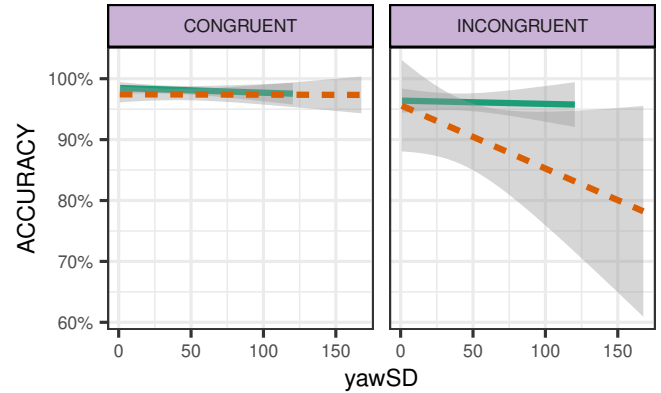


Fig. 6: Line graphs of the accuracy (y-axis) of answering congruent (left) and incongruent (right) Stroop trials by yaw standard deviation (x-axis) for participants with (solid green) and without (dashed orange) a self-avatar.

There was a significant Trial $\times$ Environment $\times$ yaw SD 3-way interaction, ($F(1, 111.97) = 5.28, p = .02, \eta^2 = .04$). See Figure 7. During the more challenging incongruent trials the stereotypical environment had a significantly more negative slope compared to the neutral environment, ($t(86.4) = 2.66, p = .009, r = .28$). In the stereotypical environment a high yaw SD correlated with low accuracy during the challenging incongruent trials. The same was not seen when participants were in the neutral environment.

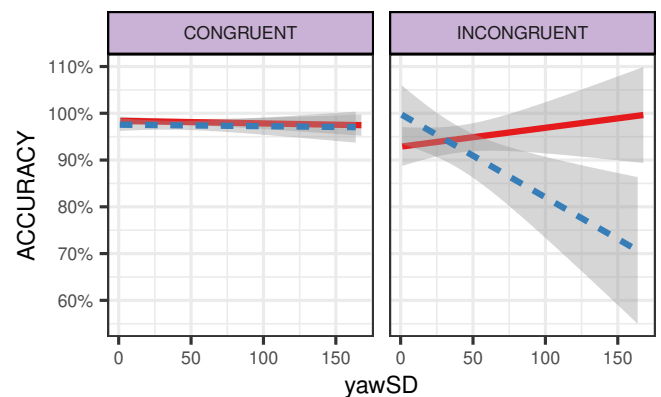


Fig. 7: Line graphs of the accuracy (y-axis) of answering congruent (left) and incongruent (right) Stroop trials by yaw standard deviation (x-axis) for participants in the neutral (solid red) and in the stereotypical (dashed blue) environments.

Finally, there was a significant Avatar $\times$ Environment $\times$ yaw SD 3-way interaction, ($F(1, 114.07) = 4.44, p = .04, \eta^2 < .01$). See Figure 8. In the stereotypical environment, participants without an avatar had a significantly more negative slope compared to participants with an avatar, ($t(105)n = 2.77, p = .007, r = .26$). Participants without

Effect	Accuracy		Latency	
	F	p	F	p
Trial	0.57	.456	19.14 ***	<.001
Env	4.16 *	.046	5.38 *	.024
Embodiment	0.37	.548	3.60 +	.068
Trial:Env	1.70	.198	0.15	.699
Trial:Embodiment	1.15	.294	0.06	.811
Env:Embodiment	4.94 *	.030	4.88 *	.032
Trial:Env:Embodiment	4.09 *	.048	0.09	.767

Table 2: Analysis of the self-avatar participants considering avatar Embodiment. Output of the Mixed Model ANOVAs for Accuracy and Latency of Stroop performance.

a self-avatar who moved their heads more had lower accuracy in the stereotypical environment compared to participants who had a self-avatar.

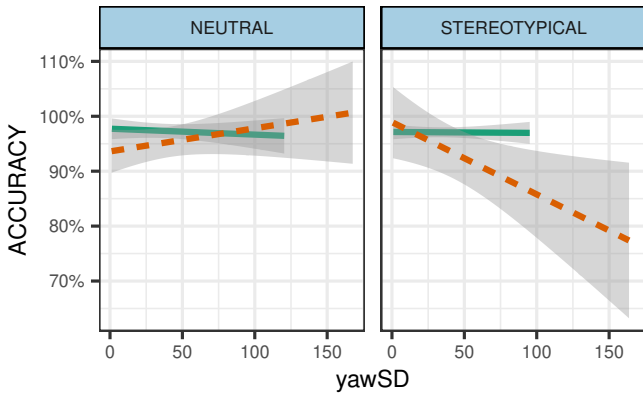


Fig. 8: Line graphs of the Stroop trial accuracy (y-axis) in the neutral (left) and stereotypical (right) environments, by yaw standard deviation (x-axis) for participants with (solid green) and without (dashed orange) a self-avatar.

The embodiment measurement is applicable for participants who had an avatar. Accuracy was analyzed for this subset of participants, with a Mixed Model ANOVA with explanatory variables Trial (In-congruent, Congruent), Environment (Neutral, Stereotypical) and Embodiment. The ANOVA output is reported in Table 2 (left).

A significant main effect of Environment was found ($F(1, 53.94) = 4.16, p < .05, \eta^2 = .07$). Participants were more accurate in the neutral environment ($M = 97.4\%, SE = .57\%$) than in the stereotypical environment ($M = 97.1\%, SE = .57\%$). The Environment main effect was moderated by a significant 2-way interaction with Embodiment, ($F(1, 54.01) = 4.94, p = .03, \eta^2 = .08$), as well as a significant Environment $\times$ Embodiment $\times$ Trial 3-way interaction ($F(1, 53.36) = 4.09, p < .05, \eta^2 = .07$).

Post-hoc analysis of the 3-way interaction was performed with estimated marginal means. See Figure 9. During the more challenging incongruent trials, the embodiment slope was significantly higher in the stereotypical environment than in the neutral environment ($t(36.6) = 2.98, p = .005, r = .44$). In the stereotype threatening environment, during the challenging incongruent trials, participants with higher subjective embodiment scores had higher accuracy. Conversely, higher embodiment in the neutral environment correlated with lower accuracy.

4.2 Latency

Latency - the time to respond on correct responses, was analyzed with a Mixed Model ANOVA with explanatory variables Trial (In-congruent, Congruent), Avatar (Avatar, No Avatar), Environment (Neutral, Stereotypical) and yaw mode during the embodiment phase. The ANOVA

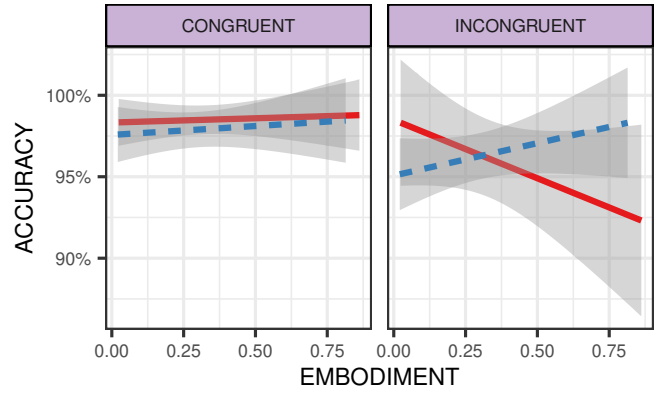


Fig. 9: Participants in the self-avatar condition: Line graphs of the accuracy (y-axis) of answering congruent (left) and incongruent (right) Stroop trials by avatar embodiment (x-axis) for participants in the neutral (solid red) and in the stereotypical (dashed blue) environments.

output is reported in Table 1 (right).

Significant main effects of Trial, Avatar, and Environment were found (See Table 1 for test statistics). Participants were slower to respond on incongruent trials ($M = 972ms, SE = 24.5$) compared to neutral trials ($M = 796ms, SE = 16.2$) with a large effect ($\eta^2 = .21$). As shown in Figure 10, participants were slower to respond without an avatar ($M = 908ms, SE = 27.2$) compared to with an avatar ($M = 860ms, SE = 27.3$) with a medium effect ($\eta^2 = .13$). Finally, as shown in Figure 11, participants were slower to respond in the stereotypical environment ($M = 896ms, SE = 20.4$) compared to the neutral environment ($M = 873ms, SE = 20.1$) with a small effect ($\eta^2 = .05$).

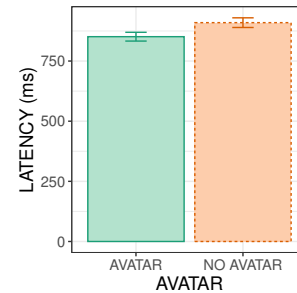


Fig. 10: A bar chart of the latency to respond on correctly answered Stroop trials for participants with (solid green) and without (dashed orange) a self-avatar.

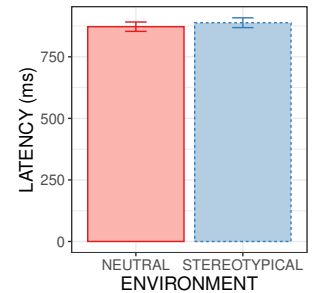


Fig. 11: A bar chart of the latency to respond on correctly answered Stroop trials for participants when they were in the neutral (solid red) and in the stereotypical (dashed blue) environments.

The Avatar and Environment main effects were moderated by interactions with yaw mode, with medium effects of $\eta^2 = .10$ and $.08$ respectively. Participants with an avatar had a steeper yaw mode slope than participants without an avatar, ($t(102) = 3.84, p < .001, r = .35$). See Figure 12. A higher yaw mode, indicating that the head spent more time in one location, had higher latency to respond. Though both axes have a positive slope, participants with an avatar had a steeper slope. However, the max yaw mode for participants with an avatar was 27.74% while the maximum yaw mode for participants without an avatar was 37.41%.

Finally, participants in the stereotypical environment with a higher yaw mode were slower to respond than participants with a smaller yaw mode. This slope was steeper than that for participants in the neutral environment, ($t(80.8) = -3.20, p = .002, r = .33$). See Figure 12.

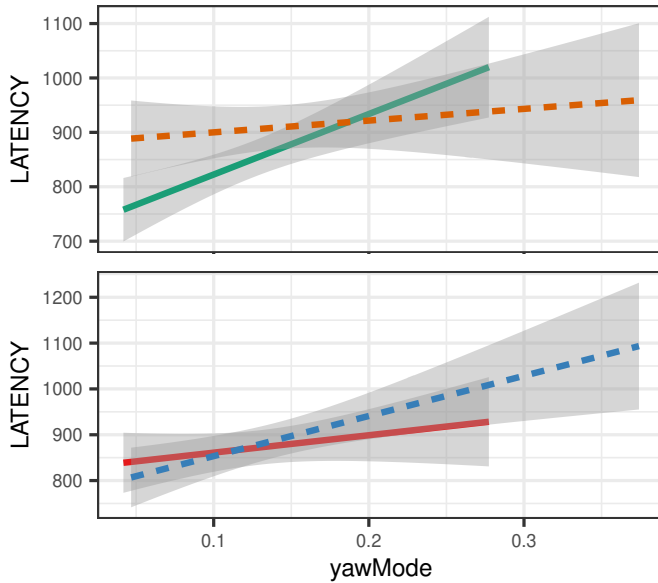


Fig. 12: Line graphs of the latency to respond in *ms* (y-axis) of Stroop trials by yaw mode (x-axis). Top: participants with (solid green) and without (dashed orange) a self-avatar. Bottom: participants in the neutral (solid red) and in the stereotypical (dashed blue) environments.

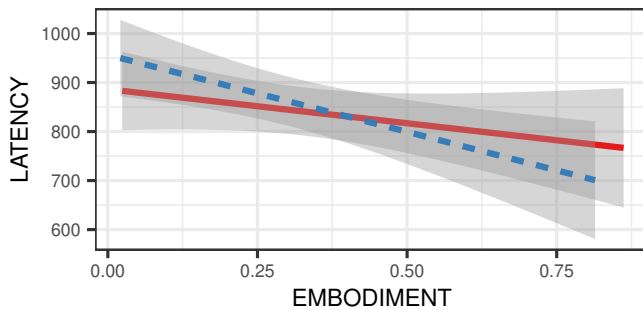


Fig. 13: Participants in the self-avatar condition: Line graphs of the latency to respond in *ms* (y-axis) of answering correct Stroop trials by avatar embodiment (x-axis) for participants in the neutral (solid red) and in the stereotypical (dashed blue) environments.

The embodiment measurement is applicable for participants who had an avatar. Latency was analyzed for this subset of participants, with a Mixed Model ANOVA with explanatory variables Trial (In-congruent, Congruent), Environment (Neutral, Stereotypical) and Embodiment. The ANOVA output is reported in Table 2 (right).

Significant main effects of Trial and Environment were found. Participants were slower to respond during incongruent trials ($M = 929ms$, $SE = 32.8$) compared to congruent trials ($M = 772ms$, $SE = 21.9$) with a large effect ($\eta^2 = .39$), and were slower to respond in the stereotypical environment ($M = 858ms$, $SE = 27.3$) compared to the neutral environment ($M = 843ms$, $SE = 27.3$) with a medium effect ($\eta^2 = .09$). Additionally, there was a significant Environment $\times$ Embodiment interaction ($F(1,54) = 4.88$, $p = .03$, $\eta^2 = .08$). Higher subjective embodiment scores correlated with faster response times with a steeper negative slope in the stereotypical environment. See Figure 13.

4.3 Questionnaire

Participants' sense of Inclusion was evaluated with a 2 (Avatar: Avatar, No Avatar) $\times$ 2 (Environment: Neutral, Stereotypical) ANOVA. Significant main effects of both Avatar ($F(1,38) = 8.73$, $p = .005$, $\eta^2 = .14$)

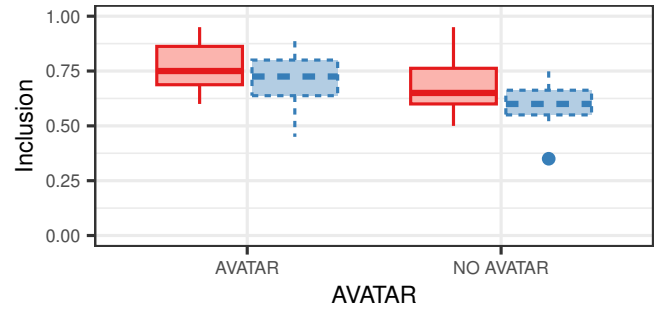


Fig. 14: Box plots of the sense of Inclusion (y-axis) for participants with (left) and without (right) a self-avatar, groups by the neutral (solid red) and the stereotypical (dashed blue) environments.

and Environment ($F(1,38) = 11.12$, $p = .005$, $\eta^2 = .08$) were found.

Participants with a self-avatar ($M = .74$, $SE = .02$) reported a higher sense of Inclusion compared to participants without an avatar ($M = .66$, $SE = .02$). Further, participants reported a significantly lower sense of inclusion in the stereotypical environment ($M = .66$, $SE = .02$) compared to the neutral environment ($M = .73$, $SE = .02$). See Figure 14.

5 DISCUSSION

Results support **H1**, that the environment will impact user performance. When women+ were in the stereotypical STEM environment and had greater head yaw SD, an indication that they were more distracted [26], and their accuracy decreased during the challenging incongruent trials. However, the same drop in accuracy was not seen when women+ were in the neutral environment regardless of increased yaw SD. Additionally, when considering latency, women+ were significantly slower to respond in the stereotypical STEM environment compared to the neutral environment. Both of these results support that the stereotypical STEM environment negatively impacted participant performance. A slower response time in the Stroop Task is a sign of working memory impairment [18] that can potentially be caused by stereotype threatening situations [27], indicating that women+ were distracted or threatened by the stereotypical STEM environment.

Additionally, there was an interaction with environment and the yaw mode, the percentage of time during the embodiment phase that a participant's head was at the mode of the yaw axis. When women+ were in the stereotypical STEM environment with greater yaw mode they were slower to respond than when they were in the neutral environment. The yaw mode is likely an indicator of focus on one location, such as a poster on the wall. The results suggest that as focus duration increased during the embodiment phase in the stereotypical environment, so did latency to respond during the testing phase. These results suggest that when asked to look around the room, women+ who focused on one point in the room, such as staring at a stereotypical STEM poster, demonstrated signs of greater working memory impairment during the testing phase compared to when they were in the neutral environment.

Replicating previous results by Cheryan et al. [7], our results support **H2**, that the environment will impact women+ subjective sense of inclusion. Women+ indicated that they felt significantly more included in the neutral environment compared to the stereotypical environment as shown by reporting significantly higher Inclusion responses in the neutral environment. Additionally, women+ felt more included when they were given a pro-STEM self-avatar. This indicates that both the environment and the avatar can induce a sense of inclusion.

Results support **H3**, that performance will be improved for participants with a self-avatar compared to participants without a self-avatar. When considering latency, participants with a self-avatar responded more quickly than participants without a self-avatar. When distracted, as indicated by having a higher yaw SD, participants in the No Avatar condition demonstrated a performance drop by being both less accurate and slower to respond compared to participants in the Avatar condition.

This phenomenon was also present when considering the environment. Participants in the stereotypical environment without a self-avatar were less accurate than participants with a self-avatar. This drop in performance occurred when participants had a higher yaw SD, an indication of distraction. Conversely, no significant differences were found in accuracy in the neutral environment. As seen in previous work [28], the avatar appeared to buffer women+ from the negative effects of the threatening environment. However, when there is no avatar the negative effects, in this case distraction and lower accuracy, were seen. Though the avatar appeared to buffer women+ from some of the negative performance drops, significant higher order effects, supporting that the performance dropped in the stereotypical environment regardless of avatar condition, were still found. This suggests that the pro-STEM avatar mitigated some of the performance drop caused by the stereotypical environment, but it may not have been able to buffer women+ completely.

Finally, results support **H4**, that women+ with higher avatar embodiment will have better performance. For women+ who had a self-avatar and a higher subjective AEQ score, differences in accuracy were seen in the more challenging incongruent trials. Higher embodiment when in the stereotypical environment was correlated with higher accuracy. Additionally, higher embodiment in the neutral environment was correlated with lower accuracy. Stereotype threat affects the out-group, in this case people who do not see themselves as identifying with the stereotypical STEM environment. Higher embodiment in the pro-STEM self-avatar may have induced a pro-STEM Proteus Effect making women+ feel like they are in the in-group. The stereotypical expectant performance for the pro-STEM avatar is to perform well. These expectant results are known to be opposite when people are not under threat (underperforming in the neutral environment) and active when under threat (performing well in the stereotypical environment) [4]. Similar performance drops were seen in the non-threatening condition for the in-group when considering gender swapping avatars and stereotype threat [28, 30].

Our results replicate previous results that small changes to a VE, i.e. changing a few posters on the walls, negatively impact women+ feeling of inclusion in a space [7–9]. Additionally, we have demonstrated that these small changes also negatively impact performance. Women+ demonstrated signs of being more distracted, were less accurate, and were slower to respond when they were in the stereotypical environment compared to the neutral environment. Of note, the posters that were selected conveyed positive or neutral messages. None of the posters were negative and similar posters could be seen in real classrooms, offices, or research labs such as those seen in Figures 15 and 16.

Previous work also supports that real world posters and items impact women's feeling of inclusion in Computer Science spaces [8, 23]. This work has demonstrated that virtual items that negatively impacted women's sense of inclusion also negatively impacted their performance. It is possible that these real world items may negatively impact performance as well and this should be investigated in future work.

5.1 Limitations and Observations

During the research, there were a few technical issues that were encountered. The HMD rebooted during the experiment for some participants. When this happened participants were instructed to remain still until it restarted, which usually took 30–60 seconds. The experimenter made notes when this happened and the experiment data collection was automatically marked in the log files and subsequently removed from analysis. Participants were told to resume when they saw the virtual environment again.

The experiment included self-avatars and participants self-selected the avatar with whom they most identified. A racially diverse collection of avatars was chosen so that participants were able to match ethnicity of their self-avatars which has been shown to enhance virtual embodiment [11]. Further, to enhance embodiment a mirror was placed on the desk in front of the participant [14]. One advantage of the mirror displayed during the experiment was that participants with an avatar were able to see themselves juxtaposed with an image in the background, which was either Albert Einstein or inclusive STEM scientists. This may have

increased the effectiveness of the environments, as these images were at the forefront of the participant's attention.

However, the mirror may have also impacted results in a different way. Two participants commented on a strategy they used during the testing phase where they focused on the text in the mirror, rather than on the screen. By looking in the mirror, these participants limited their focus on the text itself in order to prevent themselves from reading it, which helped them only see the color. This strategy may have impacted results, as the expected decrease in latency or accuracy in the incongruent trials may not have applied for these participants or others who also did the same thing. Additionally, their heads may have been more still, as they focused on a single point in order to only see the color. For future experiments using the Stroop Interference Task, the mirror on the table should not be present during the testing phase, and should only be included during an embodiment phase. The mirror may have helped a few participants to perform better than they would have without using this trick by limiting the strength of the Stroop Interference Task. Due to the counter balanced design of the experiment that all participants completed the Stroop Interference Task in both environments we do not believe this significantly impacted results.

The qualitative responses from participants also revealed interesting results. Many participants commented on the color scheme of the two environments. The stereotypical posters were much darker, as some were in black and white, while the neutral environment had more colorful and lighter posters. Future work should investigate if the colors of the posters in the environment may have impacted cognitive performance instead of the poster content. However, as this work replicated the results of Inclusion in the environment [7] it is likely that the poster content, regardless of poster color, impacted performance for women+.

5.2 Suggestions for Future Works

The results of this paper suggest that researchers conducting experiments in XR should be cautious in their experiment design, as there may be factors that influence performance in underrepresented communities including women+. Researchers should be cautious of possible biases in their experiments and strive to mitigate any variables that would cause a participant to feel unwelcome or threatened in the virtual or real environment. This applies to the physical space in which the experiment is held, as well as the designs of the virtual spaces.

In this study, all of the posters on the walls were neutral or positive, including the stereotypical STEM posters. At face value, these posters may appear to be affirmative and positive, as the Einstein and Michael Jordan posters conveyed positive messages and quotes. To a researcher, it may appear as if these are harmless posters to put up on the wall since they have positive messages. However, based on these results, they have negative gendered impacts compared to more neutral posters. Just a few masculine posters were able to negatively impact women+ performance in the experiment. Based on these results we recommend that research experiments should be designed to be as gender neutral as possible in order to minimize any women+ feeling threatened in physical or virtual environments, in order to remove external variables and increase confidence in study results. We recommend that gendered and stereotypical items should also be removed from the physical lab when conducting experiments.

Further, previous work has identified gender differences in XR experiments including, but not limited to [13, 15, 38]. However, systematic reviews have highlighted possible design bias in some studies that have found gender differences [22]. This work further supports the possibility of design bias in experiments and supports the need to replicate previous studies that have identified gender differences to determine if subtle design decisions may have unintentionally included design bias in results.

6 CONCLUSION

The results from this work support that the design of the virtual environment impacts participants. In this case, a stereotypical STEM classroom was threatening for non-male participants. Overall when



Fig. 15: A colleague wearing a Star Wars shirt during a research meeting. Photo taken with permission.



Fig. 16: Star Wars nick-knacks in a faculty office. Photo taken with permission.

women+ were in the stereotypical environment they were less accurate, slower to respond, and were more distracted. Additionally, participants who held their heads in one location longer when looking around the stereotypical room had slower response times. These same detrimental drops in performance were not seen in the neutral environment.

As predicted, the self-avatar positively impacted participants as measured by performance. The negative impact of increased head movement affected participants without a self-avatar more than participants with a self-avatar. Further, for participants with a self-avatar, higher subjective embodiment levels correlated with higher accuracy and faster response time on correct trials. Even though the self-avatar was beneficial, there were still performance differences between the two environments such that performance was worse in the stereotypical environment regardless of the self-avatar. This supports that the self-avatar was able to buffer participants from only some of the negative impacts caused by the stereotypical environments.

Stereotypical symbols are all around us as seen by a Computer Science colleague's clothing during a lab meeting (Figure 15) and nick-knacks in a faculty office (Figure 16). Similar posters or items may be located in our research labs as well. It is possible that our real world clothing, posters, and items in our labs may also impact participant performance in unknown and unexpected ways. As a community, we should consider the virtual and real environments that we are placing our participants into and carefully consider these environments in case they may be adding unintended design bias into results.

Based on these results we recommend that when designing a VE for an experiment, environments should not have more images/avatars of men than women in the scene, and stereotypical images such as well known male scientists or science fiction items should never be used. Further, we encourage researchers to remove stereotypical items from their physical labs, and wear neutral clothing when running experiments.

ACKNOWLEDGMENTS

This material is based upon work supported by the National Science Foundation under Grant 1942146. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation.

REFERENCES

- [1] B. An, F. Matteo, M. Epstein, and D. E. Brown. Comparing the performance of an immersive virtual reality and traditional desktop cultural game. In *CHIRA*, pp. 54–61, 2018. 2
- [2] D. Banakou, S. Kishore, and M. Slater. Virtually being einstein results in an improvement in cognitive task performance and a decrease in age bias. *Frontiers in psychology*, 9:368309, 2018. 3
- [3] J. Blascovich, J. Loomis, A. C. Beall, K. R. Swin, C. L. Hoyt, and J. N. Bailenson. Immersive virtual environment technology as a methodological tool for social psychology. *Psychological inquiry*, 13(2):103–124, 2002. 1
- [4] M. Cadinu, A. Maass, S. Frigerio, L. Impagliazzo, and S. Latinotti. Stereotype threat: The effect of expectancy on performance. *European journal of social psychology*, 33(2):267–285, 2003. 8
- [5] J. G. Cavazos, P. J. Phillips, C. D. Castillo, and A. J. O'Toole. Accuracy comparison across face recognition algorithms: Where are we on measuring race bias? *IEEE transactions on biometrics, behavior, and identity science*, 3(1):101–111, 2020. 1
- [6] T. M. Chaplin and A. Aldao. Gender differences in emotion expression in children: a meta-analytic review. *Psychological bulletin*, 139(4):735, 2013. 2
- [7] S. Cheryan, A. N. Meltzoff, and S. Kim. Classrooms matter: The design of virtual classrooms influences gender disparities in computer science classes. *Computers & Education*, 57(2):1825–1835, 2011. 1, 2, 3, 7, 8
- [8] S. Cheryan, V. C. Plaut, P. G. Davies, and C. M. Steele. Ambient belonging: how stereotypical cues impact gender participation in computer science. *Journal of personality and social psychology*, 97(6):1045, 2009. 2, 8
- [9] S. Cheryan, V. C. Plaut, C. Handron, and L. Hudson. The stereotypical computer scientist: Gendered media representations as a barrier to inclusion for women. *Sex roles*, 69:58–71, 2013. 2, 3, 8
- [10] A. Dirin, A. Alamäki, and J. Suomala. Gender differences in perceptions of conventional video, virtual reality and augmented reality. 2019. 2
- [11] T. D. Do, C. I. Protoko, and R. P. McMahan. Stepping into the right shoes: The effects of user-matched avatar ethnicity and gender on sense of embodiment in virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 2024. 8
- [12] T. D. Do, S. Zelenty, M. Gonzalez-Franco, and R. P. McMahan. Valid: a perceptually validated virtual avatar library for inclusion and diversity. *Frontiers in Virtual Reality*, 4, 2023. doi: 10.3389/frvir.2023.1248915 4
- [13] A. Felnhofer, O. D. Kothgassner, L. Beutl, H. Hlavacs, and I. Kryspin-Exner. Is virtual reality made for men only? exploring gender differences in the sense of presence. *Proceedings of the International Society on presence research*, pp. 103–112, 2012. 1, 2, 8
- [14] M. Gonzalez-Franco, D. Perez-Marcos, B. Spanlang, and M. Slater. The contribution of real-time mirror reflections of motor actions on virtual body ownership in an immersive virtual environment. In *2010 IEEE virtual reality conference (VR)*, pp. 111–114. IEEE, 2010. 8
- [15] S. Grassini and K. Laumann. Are modern head-mounted displays sexist? a systematic review on gender differences in hmd-mediated virtual reality. *Frontiers in psychology*, 11:1604, 2020. 1, 2, 8
- [16] S. Grassini, K. Laumann, and M. Rasmussen Skogstad. The use of virtual reality alone does not promote training performance (but sense of presence does). *Frontiers in psychology*, 11:1743, 2020. 2
- [17] A. R. Jensen and W. D. Rohwer Jr. The stroop color-word test: a review. *Acta psychologica*, 25:36–93, 1966. 3
- [18] M. J. Kane and R. W. Engle. Working-memory capacity and the control of attention: the contributions of goal neglect, response competition, and task set to stroop interference. *Journal of experimental psychology: General*, 132(1):47, 2003. 3, 7
- [19] R. Kaner and E. Frachtenberg. Experience and representation of gender minorities in undergraduate computer science. Technical report, Technical Report. EasyChair, 2020. 2
- [20] A. D. Kaplan, J. Cruik, M. Endsley, S. M. Beers, B. D. Sawyer, and P. A. Hancock. The effects of virtual reality, augmented reality, and mixed reality as training enhancement methods: A meta-analysis. *Human factors*, 63(4):706–726, 2021. 1

- [21] S. Kavanagh, A. Luxton-Reilly, B. Wuensche, and B. Plimmer. A systematic review of virtual reality in education. *Themes in science and technology education*, 10(2):85–119, 2017. 1
- [22] C. MacArthur, A. Grinberg, D. Harley, and M. Hancock. You're making me sick: A systematic review of how virtual reality research considers gender & cybersickness. In *Proceedings of the 2021 CHI conference on human factors in computing systems*, pp. 1–15, 2021. 2, 8
- [23] A. Master, S. Cheryan, and A. N. Meltzoff. Computing whether she belongs: Stereotypes undermine girls' interest and sense of belonging in computer science. *Journal of educational psychology*, 108(3):424, 2016. 2, 8
- [24] A. Master, A. N. Meltzoff, and S. Cheryan. Gender stereotypes about interests start early and cause gender disparities in computer science and engineering. *Proceedings of the National Academy of Sciences*, 118(48):e2100030118, 2021. 2
- [25] C. Mousas, D. Anastasiou, and O. Spantidi. The effects of appearance and motion of virtual characters on emotional reactivity. *Computers in Human behavior*, 86:99–108, 2018. 2
- [26] A. Narayanan, R. M. Kaimal, and K. Bijlani. Estimation of driver head yaw angle using a generic geometric model. *IEEE Transactions on Intelligent Transportation Systems*, 17(12):3446–3460, 2016. 3, 7
- [27] J. W. Osborne et al. Gender, stereotype threat, and anxiety: Psychophysiological and cognitive evidence. 2006. 3, 7
- [28] T. C. Peck, M. Doan, K. A. Bourne, and J. J. Good. The effect of gender body-swap illusions on working memory and stereotype threat. *IEEE transactions on visualization and computer graphics*, 24(4):1604–1612, 2018. 3, 8
- [29] T. C. Peck and M. Gonzalez-Franco. Avatar embodiment: a standardized questionnaire. *Frontiers in Virtual Reality*, 1:575943, 2021. 3, 5
- [30] T. C. Peck, J. J. Good, and K. A. Bourne. Inducing and mitigating stereotype threat through gendered virtual body-swap illusions. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, pp. 1–13, 2020. 3, 8
- [31] T. C. Peck, K. A. McMullen, and J. Quarles. Divrsify: Break the cycle and develop vr for everyone. *IEEE Computer Graphics and Applications*, 41(6):133–142, 2021. 1
- [32] T. C. Peck, S. Seinfeld, S. M. Aglioti, and M. Slater. Putting yourself in the skin of a black avatar reduces implicit racial bias. *Consciousness and cognition*, 22(3):779–787, 2013. 3
- [33] T. C. Peck, L. E. Sockol, and S. M. Hancock. Mind the gap: The underrepresentation of female participants and authors in virtual reality research. *IEEE transactions on visualization and computer graphics*, 26(5):1945–1954, 2020. 1
- [34] T. C. Peck and A. Tutar. The impact of a self-avatar, hand collocation, and hand proximity on embodiment and stroop interference. *IEEE transactions on visualization and computer graphics*, 26(5):1964–1971, 2020. 2, 3, 4
- [35] L. Phillips, B. Ries, M. Kaeding, and V. Interrante. Avatar self-embodiment enhances distance perception accuracy in non-photorealistic immersive virtual environments. In *2010 IEEE virtual reality conference (VR)*, pp. 115–1148. IEEE, 2010. 2
- [36] L. D. Pittman and A. Richmond. Academic and psychological functioning in late adolescence: The importance of school belonging. *The Journal of Experimental Education*, 75(4):270–290, 2007. 2
- [37] M. S. Porta, S. Greenland, M. Hernán, I. dos Santos Silva, and J. M. Last. *A dictionary of epidemiology*. Oxford University Press, USA, 2014. 1
- [38] V. Schwind, P. Knierim, C. Tasci, P. Franczak, N. Haas, and N. Henze. "these are not my hands!" effect of gender on the perception of avatar hands in virtual reality. In *Proceedings of the 2017 CHI conference on human factors in computing systems*, pp. 1577–1582, 2017. 2, 8
- [39] K. Stanney, C. Fidopiastis, and L. Foster. Virtual reality is sexist: but it does not have to be. *Frontiers in Robotics and AI*, 7:476417, 2020. 1
- [40] A. Steed, Y. Pan, F. Zisch, and W. Steptoe. The impact of a self-avatar on cognitive load in immersive virtual reality. In *2016 IEEE virtual reality (VR)*, pp. 67–76. IEEE, 2016. 2
- [41] C. M. Steele. Stereotyping and its threat are real. 1998. 3
- [42] A. D. Sterling, M. E. Thompson, S. Wang, A. Kusimo, S. Gilmartin, and S. Sheppard. The confidence gap predicts the gender pay gap among stem graduates. *Proceedings of the National Academy of Sciences*, 117(48):30303–30308, 2020. 2
- [43] E. A. Suma, S. L. Finkelstein, S. Clark, P. Goolkasian, and L. F. Hodges. Effects of travel technique and gender on a divided attention task in a virtual environment. In *2010 IEEE Symposium on 3D User Interfaces (3DUI)*, pp. 27–34. IEEE, 2010. 2
- [44] A. S. Won, B. Perone, M. Friend, and J. N. Bailenson. Identifying anxiety through tracked head movements in a virtual classroom. *Cyberpsychology, Behavior, and Social Networking*, 19(6):380–387, 2016. 3
- [45] N. Yee and J. Bailenson. The proteus effect: The effect of transformed self-representation on behavior. *Human communication research*, 33(3):271–290, 2007. 2